YOLO with Attention Mechanisms for Building Cracks Detection

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Abstract

This study proposes an improved YOLO11 model for crack detection by removing the C2PSA module and integrating attention mechanisms in the Neck. A composite crack dataset was used for training and testing. Experimental results show that the proposed model outperforms YOLO11n and YOLO8n in both mAP@0.5 and mAP@0.5:0.95. The findings confirm that the model effectively enhances fine crack detection under limited computational resources and shows potential for deployment on edge devices to improve building safety.

1 Introduction

Insufficient infrastructure inspection poses risks to public safety, while manual inspection is constrained by efficiency and cost. With the advancement of deep learning and computer vision, the YOLO series has gained attention for its real-time performance and accuracy; however, the C2PSA module in YOLO11 [1] remains constraint in extracting fine features. This study proposes an enhanced YOLO11 model by removing C2PSA and incorporating three attention modules. Experimental results show that the proposed model outperforms YOLO11n and YOLO8n in both mAP@0.5 and mAP@0.5:0.95, while retaining lightweight efficiency.

2 Methodology

The proposed model adopts a modified YOLO11 architecture for crack detection in building and bridge structures, as illustrated in Fig. 1. The C2PSA module in the original YOLO11 is removed, as it may interfere with other attention mechanisms and add unnecessary computational overhead. Inspired by the YOLOv8-AM design [2], three attention modules —Global Attention Mechanism (GAM) [3], Residual Convolutional Block Attention Module (Res CBAM) [4][5] and Shuffle Attention (SA) [6]— are individually inserted into the Neck to strengthen the representation of fine crack features. This modification enables the model to better capture slender and low-contrast cracks while suppressing background interference. The Head retains the original YOLO11 design, with BCE for classification and CIoU+DFL for bounding box regression.

The proposed model was trained and evaluated on a composite dataset constructed from five publicly available crack datasets: BAC_HIEN_Crack_Concrete_2024 [7], Crack Detection.v2 [8], Crack Detection.v3i [9], Crack Finder.v1i [10], and the Concrete Crack Images for Classification dataset [11]. The combined dataset is divided into 15,524 training images, 2,276 validation images, and 1,665 test images.All experiments are conducted using PyTorch 2.0 with CUDA 12.2 on an NVIDIA RTX 4090 GPU and an Intel(R) Core (TM) i9-14900K CPU (3.20 GHz) with 64 GB RAM.

3 Result

To validate the effectiveness of the proposed model, comparisons were made with YOLO11n and YOLO8n, with evaluation carried out in terms of model complexity, accuracy, and inference speed.

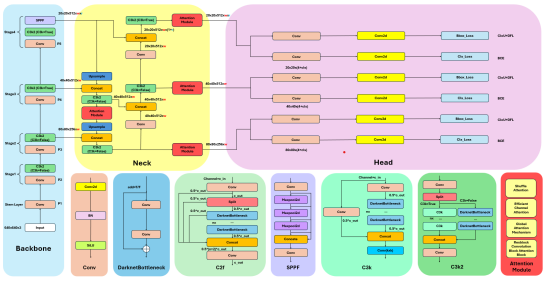


Fig. 1: YOLO11 variant with attention modules integrated into the Neck (full-size diagram in Appendix A).

3.1 Tables

Table 1 Comparison of model complexity

Model	Layers	Parameters	Gradients	Flops
GAM	333	2898515	2898499	7.2×10^9
ResBlock CBAM	365	3450139	3450123	8.3×10^9
SA	305	2217451	2217435	5.8×10^9
YOLO11n	181	2590035	2590019	6.4×10^9
YOLO8n	225	3011043	3011027	8.2×10^9

3.1.1 Model Complexity: Table 1 compares the models in terms of parameters and FLOPs. Both GAM and SA are close to YOLO11n in parameter size, with SA requiring only 2.22M, even lower than YOLO11n (2.59M), while ResBlock CBAM is slightly more (3.45M,). In terms of FLOPs, SA (5.8×10^9) and GAM (7.2×10^9) remain comparable to YOLO11n (6.4×10^9), whereas ResBlock CBAM is closer to YOLO8n. These results indicate that the proposed variants maintain lightweight complexity without significant computational overhead, making them suitable for real-time crack detection.

Table 2 Comparison of model accuracy

Model	map50	map50:95	precision	recall
GAM	0.9912	0.9481	0.9849	0.9855
ResBlock CBAM	0.9904	0.9451	0.9843	0.9845
SA	0.9899	0.9392	0.9832	0.9835
YOLO11n	0.9833	0.9112	0.9717	0.9720
YOLO8n	0.9707	0.8921	0.9738	0.9480

3.1.2 Accuracy: Table 2 compares accuracy of various variants. GAM achieved the best results across all metrics, while ResBlock

CBAM and SA also outperforms YOLO11n and YOLO8n. Overall, the three variants show clear accuracy gains, confirming the benefit of attention mechanisms.

Table 3 Comparison of inference speed

Model	Image Count	Average Time (s)	Total Time (s)	FPS
GAM	176	0.0716	12.5985	13.97
ResBlock CBAM	176	0.0684	12.0343	14.62
SA	176	0.0673	11.8485	14.85
YOLO11n	176	0.0676	11.9060	14.78
YOLO8n	176	0.0737	12.9699	14.78

3.1.3 Inference Speed: Table 3 compares inference performance. SA achieves the highest FPS (14.85), GAM and ResBlock CBAM are similar to YOLO11n, while YOLO8n is the slowest (13.57 FPS). Although the attention modules may add computation, the impact on inference is minimal, and real-time capability is maintained.

4 Conclusion

This research work proposes a modified YOLO11 for building crack detection, removing C2PSA and adding GAM, ResCBAM, and SA, respectively. The model outperforms YOLO11n and YOLO8n in mAP@0.5 and mAP@0.5:0.95, while preserving real-time inference.

5 Acknowledgements

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6 References

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7 Appendix

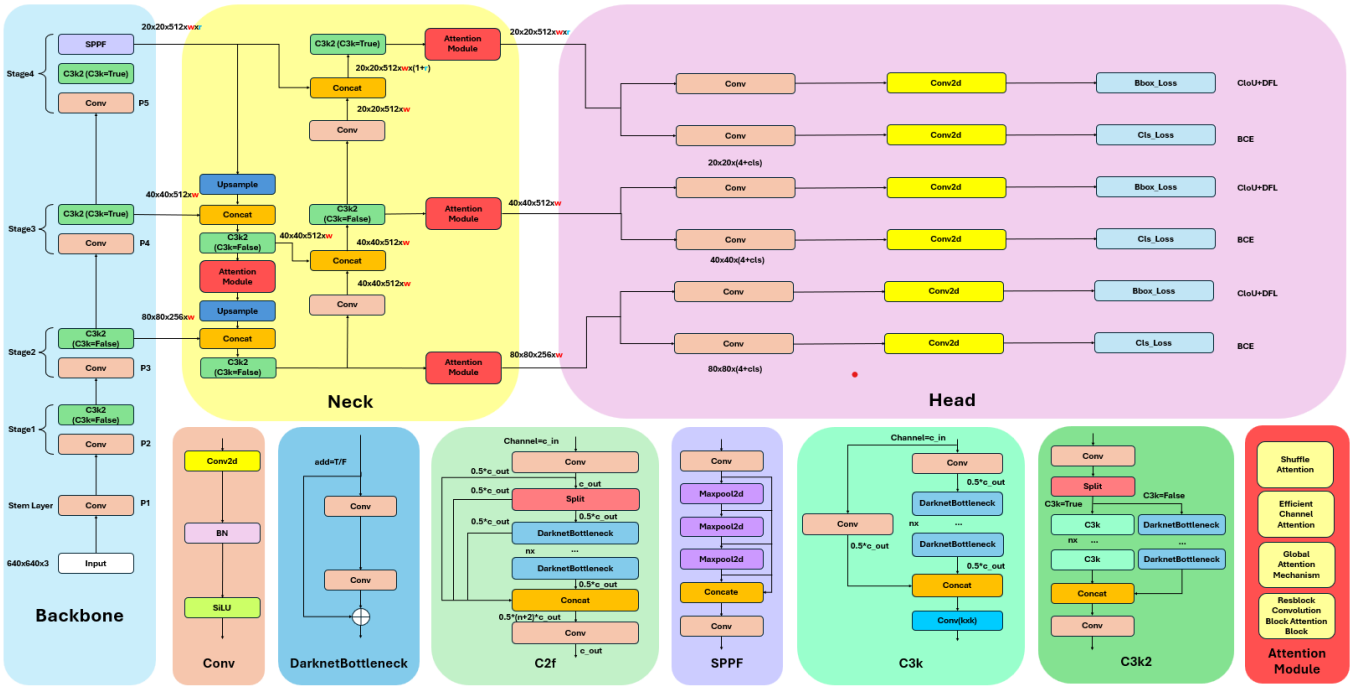


Fig. 2: Appendix A. YOLO11 variant with attention modules.